Semester Project (Sample Project)

Course 602 – Data Analytics using Python

Project Title:

Exploratory Data Analysis and Sales Prediction using Python

Dataset Details (Open Source)

[Student group size : maximum 2 (Individual project preferable)

Data source : Students are expected to use and download relevant dataset from open data source like Kaggle, UCI etc.]

Project definition : Given by the Institute / student select it but approved by their faculty member.

by: Dr. Snehal K Joshi

Mode of Project : In-house in lab / Using their own Lap-top

Dataset Name: Superstore Sales Dataset (example)
Source: Kaggle / UCI database / any other Open Dataset
Dataset Link: Provide dataset link from where the dataset is downloaded. (e.g. *link of downloaded dataset from Kaggle)*
Data Type: Structured CSV dataset
Domain: Retail / Business Analytics / healthcare etc.

Dataset Description

The dataset contains information about:

- Order details
- Sales, profit, discount
- Product categories and sub-categories
- Customer and regional information
- Dates and quantities

This dataset is ideal for:

- EDA (Univariate, Bivariate, Multivariate)
- Handling missing values and outliers
- Regression analysis
- Introduction to supervised learning

Project Objectives

After completing this project, students will be able to:

1. Understand and explore real-world datasets using Python
2. Perform **Exploratory Data Analysis (EDA)**
3. Identify data types and data distributions
4. Handle missing values and outliers
5. Automate EDA using Python libraries
6. Apply **basic regression techniques**
7. Understand supervised learning concepts
8. Evaluate model performance using error metrics

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Python Tools & Libraries to be Used:

- Python 3.x
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Project Tasks:

Task 1: Data Understanding

- Load the dataset using Pandas
- Display:
  - First 5 rows

    - Last 5 rows
    - Dataset shape
    - Column names
- Identify:
    - Quantitative data (Discrete & Continuous)
    - Qualitative data (Nominal & Ordinal)

Task 2: Exploratory Data Analysis (EDA)

Univariate Analysis

- Analyze:
    - Sales
    - Profit
    - Quantity
- Use:
    - Histogram
    - Boxplot

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Bivariate Analysis

- Sales vs Profit
- Discount vs Profit
- Quantity vs Sales
- Use:
    - Scatter plots
    - Correlation matrix

Multivariate Analysis

- Sales, Profit, Discount across Categories
- Use:
    - Pair plots
    - Heatmap

Task 3: Handling Missing Data and Outliers

- Identify missing values
- Handle missing values using:
  - Mean / Median / Mode
- Detect outliers using:
  - Boxplots
- Explain the impact of outliers

Task 4: Spread of Data

- Check data distribution:
  - Normal distribution
  - Skewed distribution
- Calculate:
  - Mean
  - Median
  - Standard deviation
  - Skewness
  - Kurtosis
- Interpret results

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Task 5: Automating EDA using Python

- Use Pandas and NumPy functions:
  - describe()
  - info()
  - isnull()
  - corr()
- Write reusable Python functions for EDA

Task 6: Regression Analysis

- Identify:
  - Dependent variable (Sales)
  - Independent variables (Quantity, Discount)
- Perform:
  - Simple Linear Regression

- Analyze:
  - Covariance
  - Correlation

Task 7: Supervised Learning – Regression Model

- Split dataset into:
  - Training data
  - Testing data
- Build regression model using:
  - LinearRegression (sklearn)
- Explain:
  - Overfitting
  - Underfitting

- Train a regression model using training data
- Validate model performance using validation data
- Test final model on test data
- Build :
  - (i) Simple Linear Regression Model
  - (ii) Multi Linear Regression Model
  - (iii) Logistic Regression Model

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Task 9: Overfitting and Underfitting Analysis

- Explain concepts of overfitting and underfitting
- Compare training and testing errors
- Demonstrate how model complexity affects performance

Task 10: Classification Task

- Convert the problem into a classification problem (Example: Profit → Profitable / Not Profitable)
- Build a classification model using supervised learning
- Evaluate classification performance using:
- Accuracy

- Confusion Matrix

Task 11: Model Evaluation

- Calculate and interpret:
  - Mean Squared Error (MSE)
  - Mean Absolute Error (MAE)
  - R^2 Score (for regression)
- Compare regression and classification outcomes

Task 12: Model Evaluation

- Calculate:
  - Mean Squared Error (MSE)
  - Mean Absolute Error (MAE)
  - R^2 Score
- Interpret model performance

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Task 13: Data visualization

- Plots:
  - Univariate analysis plots
  - Bi-variate analysis plots
  - Multi-variate analysis plots
- Data normalization plots (Normal distribution, skewness)

Expected Project Outcomes

- Perform complete EDA on real data
- Understand data behavior using visualizations
- Apply regression models
- Evaluate models using error metrics
- Connect theory with practical implementation

Project Submission Guidelines

- Python Notebook (.ipynb)
- Project Report (Word / PDF) containing:
  - Introduction
  - Dataset description
  - Analysis and outputs
  - Observations and conclusion
- Screenshots of graphs and results

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Suggested Evaluation Criteria (Internal/External):

Component	Marks
Data Understanding & Cleaning	20
EDA & Visualization	25
Regression & ML Concepts	25
Model Evaluation	15
Report & Presentation	15
Total	**100**